

CONSTRUCTION TECHNOLOGY

Construction technology has always played a central role in society. It is a multifaceted industry in which projects can result in products, systems, processes, or services and encompass various aspects of production, repair, and maintenance. In construction technology courses, students will learn about designing, constructing, and maintaining a variety of buildings and structures, and will gain experience with the tools, equipment, and processes commonly used in the field. Students will also learn about health and safety standards in the construction industry, building codes and regulations, and employment opportunities and careers in the various sectors of the industry, including electricity, carpentry, masonry, heating and cooling, and plumbing.

The construction technology courses initially focus on residential and light construction systems related to residential buildings and structures, and progress to more advanced residential and light commercial construction. The woodworking courses allow students to concentrate on furniture building or cabinet making.

The list of approved emphasis areas for construction technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Construction Engineering Technology, Grade 12

College Preparation

TCJ4C

This course enables students to further develop knowledge and skills related to residential construction and to explore light commercial construction. Students will gain hands-on experience using a variety of materials, processes, tools, and equipment, and will learn more about building design and project planning. They will continue to create and interpret construction drawings and will extend their knowledge of construction terminology and of relevant building codes and regulations, as well as health and safety standards and practices. Students will also focus on environmental and societal issues related to construction engineering technology, and will explore career opportunities in the field.

Prerequisite: Construction Engineering Technology, Grade 11, College Preparation

A. CONSTRUCTION TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of natural and manufactured materials, construction processes, and building components;
- A2.** demonstrate an understanding of building codes, regulations, and standards that govern residential and light commercial construction projects;
- A3.** demonstrate an understanding of the systems in residential and light commercial buildings;
- A4.** demonstrate an understanding of design considerations for residential and light commercial buildings;
- A5.** use construction terminology correctly.

SPECIFIC EXPECTATIONS

A1. Materials, Processes, and Components

By the end of this course, students will:

- A1.1** assess the properties (*e.g., physical, mechanical, thermal*) of various natural and manufactured building materials (*e.g., wood studs, steel studs, laminated beams, engineered floor joists*);
- A1.2** compare different building materials and techniques that can be used for the same purpose (*e.g., cement block versus poured concrete for a foundation, wood versus steel doors, digging a trench versus using a pneumatic torpedo to install a water service, nails versus adhesive for attaching panelling*);
- A1.3** describe the various processes used to produce common construction materials and components (*e.g., concrete and masonry products, engineered wood products, doors, windows, trusses*);
- A1.4** compare the processes for constructing common types of foundations and structural supports for a building (*e.g., slip forming, insulated concrete forming, post and beam, platform framing, balloon framing*);
- A1.5** assess various types of structural and non-structural components of a building (*e.g., structural: floors, studs, trusses, rafters, engineered floor joists, laminated beams; non-structural: trim, stiles, cabinetry, curtain walls, siding*);

- A1.6** describe the relationship between structural and architectural components in a building (*e.g., cladding on a support pier, material concealing a steel beam*).

A2. Building Codes, Regulations, and Standards

By the end of this course, students will:

- A2.1** identify and explain the building codes that apply to residential and light commercial construction (*e.g., Ontario Building Code, Ontario Electrical Safety Code, Ontario Fire Code*);
- A2.2** identify the authorities that regulate residential and light commercial construction (*e.g., municipalities, conservation authorities*), and explain the pertinent regulations (*e.g., requirements for setbacks, easements, and barrier-free access*);
- A2.3** identify permits and inspections required for residential and light commercial construction projects (*e.g., building, plumbing, electrical*);
- A2.4** identify and describe the standards that apply to residential and light commercial construction projects (*e.g., standards from the Workplace Safety and Insurance Board [WSIB], the Canadian Standards Association [CSA], and Underwriters Laboratories of Canada [ULC]*);
- A2.5** describe standards for sustainable building practices (*e.g., Leadership in Energy and Environmental Design [LEED®] certification standards*).

A3. Building Systems

By the end of this course, students will:

A3.1 describe the foundations of typical residential and light commercial buildings, and outline the requirements of the Ontario Building Code that apply to these foundations;

A3.2 describe the structural systems of typical residential and light commercial buildings, and outline the requirements of the Ontario Building Code that apply to these systems;

A3.3 describe the components of electrical systems of typical residential and light commercial buildings (*e.g., service, circuit-breaker panels, wiring, fixtures*), and outline the requirements of the Ontario Electrical Safety Code that apply to these systems;

A3.4 describe the components of plumbing systems of typical residential and light commercial buildings (*e.g., water supply, piping, fixtures, sewer or septic tank connection*), and outline the requirements of the Ontario Building Code (Parts 7 and 8) that apply to these systems;

A3.5 describe heating and cooling systems in typical residential and light commercial buildings (*e.g., forced-air furnace, hot-water boiler, ducts, electrical heating, air conditioning*), and outline the requirements of the Ontario Building Code (Part 6) that apply to these systems.

A4. Design Considerations

By the end of this course, students will:

A4.1 assess the human factors (*e.g., function, ergonomics, aesthetics*) to consider when planning aspects of a building (*e.g., size, appearance, traffic flow, means of egress, use of non-allergenic materials*);

A4.2 explain the reasons for selecting a particular structure and materials for a construction project (*e.g., structure: wood versus steel beams; materials: asphalt or cedar shingles versus steel roofing, siding versus brick*);

A4.3 describe factors affecting the design of a foundation (*e.g., drainage, soil type, load, frost penetration*);

A4.4 describe the weather-related loads and stresses that a building must be designed to withstand (*e.g., force of wind, snow load on roofs, expansion and contraction due to changes in temperature and humidity*);

A4.5 compare various building types (*e.g. residential, light commercial*), architectural styles (*e.g., Colonial, Tudor, Victorian, Aboriginal*), and their engineering features (*e.g., engineered floor joists, exposed roof trusses, steel beams, fire barriers*).

A5. Terminology

By the end of this course, students will:

A5.1 use correct terminology to identify and describe construction components and materials (*e.g., stud, girder, ground-fault circuit interrupter, P-trap, plenum, conduit, armoured cable*);

A5.2 use correct terminology to identify and describe construction processes and concepts (*e.g., processes: dress, rip, crosscut, joint; concepts: slope, load, thermal resistance, stability, compression strength*);

A5.3 use correct terminology to identify and describe construction tools and equipment (*e.g., air compressor, laser level, backhoe, framing hammer, wire stripper, tri-square*);

A5.4 use correct terminology to identify and describe the documents used for planning, scheduling, and controlling the construction process (*e.g., schedules, contracts, Gantt charts, permits, sub-trade contracts*).

B. DESIGN, LAYOUT, AND PLANNING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** apply a design process, other problem-solving techniques, and related concepts and principles, as appropriate, to plan construction projects and develop solutions for construction problems and challenges;
- B2.** create and interpret drawings of residential and light commercial construction projects;
- B3.** determine, use, and communicate accurate technical data for construction projects;
- B4.** plan systems for residential and/or light commercial buildings;
- B5.** apply the mathematical skills required in designing, laying out, and preparing estimates for residential and light commercial construction projects.

SPECIFIC EXPECTATIONS

B1. Design and Planning

By the end of this course, students will:

- B1.1** explain how the design process is used in the construction industry (*e.g., developing floor plans, designing prefabricated buildings; in urban planning*);
- B1.2** use appropriate problem-solving processes and techniques to address challenges related to various residential and light commercial construction projects;
- B1.3** apply technological concepts (*e.g., aesthetics, ergonomics, function*) to the design of construction projects (*e.g., shape of staircases, appearance of facades, layout of floors, placement of electrical and plumbing fixtures*);
- B1.4** apply design principles and elements to the design of construction projects (*e.g., principles: scale, proportion, balance, contrast; elements: shape, space, line, texture, colour*).

B2. Technical Drawings

By the end of this course, students will:

- B2.1** interpret technical drawings to accurately plan and lay out residential and light commercial construction projects (*e.g., determine dimensions, materials required, hardware placement, window and door sizes and types, and location of electrical and plumbing fixtures*);

- B2.2** create at least one working drawing of a building system (*e.g., structural, electrical, plumbing, heating/ventilation/air-conditioning*) with manual and/or computer-assisted methods, using appropriate views (*e.g., orthographic, plan, elevation, isometric, 3D, section, detail*) and metric and/or imperial units.

B3. Using Technical Data

By the end of this course, students will:

- B3.1** use various resources to determine technical data, code requirements, and standards for construction projects (*e.g., reference charts and tables; reports; published codes, regulations, and standards; guides and trade manuals; manufacturers' instructions; government and association websites*);
- B3.2** describe the building code requirements for forces and stresses due to loads and weather conditions (*e.g., heat, cold, wind, snow, rain*) that structural members (*e.g., footings, foundations, floors, bearing walls, roofs, columns, beams, lintels*) must be designed to withstand;
- B3.3** communicate accurate technical information verbally and graphically.

B4. Building Systems

By the end of this course, students will:

- B4.1** develop plans for a foundation suitable for the site conditions and building design (*e.g., site conditions: hydrostatic pressure, drainage, soils, frost penetration; design: type of framing and exterior wall finishes*);
- B4.2** plan the installation of the systems used in typical residential and/or light commercial buildings (*e.g., foundation, framing, electrical, plumbing, heating/ventilation/air-conditioning*);
- B4.3** identify and compare modifications (*e.g., upgraded ventilation, insulation, or air barriers*) that would improve the quality and value of a building project (*e.g., resale value, cost recovery*).

B5. Mathematical Skills

By the end of this course, students will:

- B5.1** calculate dimensions and lay out construction projects using relevant mathematical principles and formulas (*e.g., Pythagorean theorem, volume and area formulas*);
- B5.2** convert between fractions and decimals and between metric and imperial units, using appropriate tables, charts, software, and/or online conversion tools;
- B5.3** use the tables in Part 9 of the Ontario Building Code to determine the required sizes, spacing, and number of structural components for construction projects (*e.g., footings, floor joists, wall studs, rafters, beams, columns*);
- B5.4** prepare detailed, accurate estimates of the quantities and costs of materials required for construction projects (*e.g., concrete, aggregate, reinforcing steel, asphalt, lumber, roofing, interior and exterior finishes*).

C. FABRICATION, ASSEMBLY, AND FINISHING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate appropriate technical skills, including the safe use of construction tools, equipment, and materials;
- C2.** demonstrate safe and accurate building techniques;
- C3.** apply various finishes to complete residential and light commercial construction projects.

SPECIFIC EXPECTATIONS

C1. Technical Skills

By the end of this course, students will:

- C1.1** demonstrate safe work practices when using hand and power tools, materials, and equipment;
- C1.2** use metric and imperial units correctly to measure and specify materials and components for a variety of construction projects;
- C1.3** lay out construction projects using a variety of tools and equipment (*e.g., framing square, laser level, string line, plumb bob*).

C2. Fabrication and Assembly

By the end of this course, students will:

- C2.1** construct projects in accordance with design specifications (*e.g., working drawings, lists of specified materials and fixtures*);
- C2.2** safely install various systems of a construction project (*e.g., electrical, plumbing, heating/ventilation/air-conditioning*) in accordance with codes, regulations, and standards (*e.g., Ontario Building Code requirements for joists and beams*);
- C2.3** describe and implement appropriate solutions and/or modifications in response to problems encountered in residential and light commercial construction projects.

C3. Finishing

By the end of this course, students will:

- C3.1** apply appropriate exterior materials to finish residential and light commercial construction projects (*e.g., siding, brick, stone, exterior doors, windows, metal roofing*);
- C3.2** apply appropriate interior materials to finish residential and light commercial construction projects (*e.g., baseboard, wainscoting, crown moulding, interior doors, hardware, built-ins*).

D. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** identify and evaluate measures that can be taken to conserve resources on construction projects;
- D2.** explain how the construction industry and society affect each other.

SPECIFIC EXPECTATIONS

D1. Technology and the Environment

By the end of this course, students will:

- D1.1** plan projects and construction processes to minimize waste (*e.g., use efficient cutting patterns, reuse leftover material*);
- D1.2** describe the costs and benefits of environmentally friendly building practices (*e.g., high-efficiency heating and cooling, renewable energy technologies, reuse of grey water, use of materials produced from sustainable resources*);
- D1.3** compare ways of reducing the environmental footprint of construction projects through the choice of energy sources (*e.g., solar, geothermal, wind*), building design (*e.g., extra insulation, high-efficiency heating systems, green roof*), and construction processes (*e.g., use of recycled material, fuel-efficient equipment*);
- D1.4** outline strategies to reduce, reuse, and recycle construction materials, and identify methods for implementing sustainable building practices (*e.g., work with a community partner or government agency, help develop local programs, create a public awareness campaign*);
- D1.5** research and describe strategies for implementing sustainable building practices (*e.g., Canada Green Building Council guidelines, Forest Stewardship Council Canada standards*).

D2. Technology and Society

By the end of this course, students will:

- D2.1** research and assess the economic and social effects of the construction industry (*e.g., creation of primary and secondary jobs, transport of materials, land use, resource management, encroachment on Aboriginal lands*);
- D2.2** describe the factors affecting the quality of life of the occupants of residential and/or light commercial buildings (*e.g., air quality, allergens, carcinogens, aesthetics, access to transit and other services*);
- D2.3** describe how provincial and municipal regulations affect the design and cost of construction projects and influence how the construction industry meets societal needs;
- D2.4** identify factors to consider in community planning (*e.g., population density, ecology, culture*).

E. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** demonstrate an understanding of and comply with health and safety regulations and practices specific to the construction industry;
- E2.** demonstrate an understanding of careers in the construction industry and the education, training, and workplace skills required for these careers.

SPECIFIC EXPECTATIONS

E1. Health and Safety

By the end of this course, students will:

- E1.1** assess hazards related to construction materials, processes, tools, and equipment (*e.g., toxic or flammable fumes from solvents, paints, varnishes, and gasoline; explosion or burns from propane; lung damage from silica; tripping or falls in unfinished buildings; shock from damaged power tools or electrical equipment*), and describe the precautions that should be taken to avoid these hazards;
- E1.2** describe and comply with health and safety legislation and practices for the construction industry (*e.g., Workplace Safety and Insurance Board [WSIB] regulations, provincial labour legislation, local by-laws*);
- E1.3** use, handle, and store materials in accordance with Workplace Hazardous Materials Information System (WHMIS) guidelines;
- E1.4** describe the rights and responsibilities of employees (*e.g., the right to know, the right to refuse, the right to participate, as outlined in the Occupational Health and Safety Act*);
- E1.5** demonstrate the understanding of when and how to use appropriate protective clothing, gear, and equipment (*e.g., hard hat, respirator, safety harness*).

E2. Career Opportunities

By the end of this course, students will:

- E2.1** describe a variety of careers in the construction industry (*e.g., contractor, architect, engineer, tradesperson, technician, technologist, labourer, project manager*), and identify relevant postsecondary programs and their admission requirements;
- E2.2** explain the importance of lifelong learning for careers in the construction industry;
- E2.3** demonstrate an understanding of and apply the Essential Skills that are important for success in the construction industry, as identified in the Ontario Skills Passport (*e.g., computer use, data analysis, measurement and calculation*);
- E2.4** demonstrate an understanding of and apply the work habits that are important for success in the construction industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, initiative*);
- E2.5** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in construction technology (*e.g., Passport to Safety certificate, technical drawings, reports, photographs of projects, Ontario Skills Passport Work Plan and Transition Plan*), and explain why having a current portfolio is important for career development and advancement.